

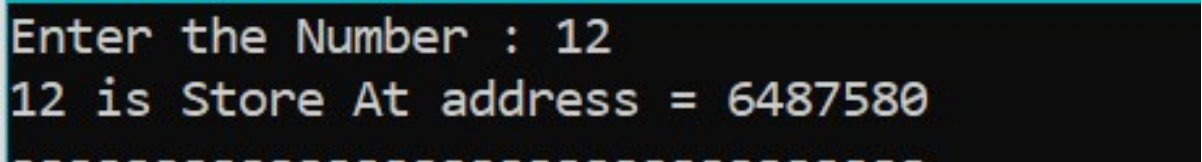
Practical-7: Basic Programs Using Pointer

1) Print value and address of a variable. (A)

Program :-

```
#include <stdio.h>
void main(){
    int a = 0;
    printf("Enter the Number : ");
    scanf("%d",&a);
    printf("%d is Store At address = %d",a,&a);
}
```

Output :-



```
Enter the Number : 12
12 is Store At address = 6487580
```

2) Demonstrate int, float, double and char pointer.(A)

Program :-

```
#include <stdio.h>
void main(){
    int a = 10;
    float b = 12.34;
    double c = 123.4567;
    char d = 'd';
    int *p1;//it can store only int data variable address
    float *p2;
    double *p3;
    char *p4;
    p1 = &a;
    p2 = &b;
    p3 = &c;
    p4 = &d;
    printf("int pointer -> %d \n",*p1);
    printf("Float pointer -> %f \n",*p2);
    printf("Double pointer -> %lf \n",*p3);
    printf("char pointer -> %c",*p4);
}
```

Output :-

```
int pointer -> 10
Float pointer -> 12.340000
Double pointer -> 123.456700
char pointer -> d
```

3) Calculate sum of two numbers using pointer. (A)

Program :-

```
#include <stdio.h>
void main(){
    int n1,n2;
    printf("Enter the First Number : ");
    scanf("%d",&n1);
    printf("Enter the Second Number : ");
    scanf("%d",&n2);
    int *p1,*p2;
    p1 = &n1;
    p2 = &n2;
    printf("%d + %d = %d",*p1,*p2,(*p1 + *p2));
}
```

Output :-

```
Enter the First Number : 12
Enter the Second Number : 34
12 + 34 = 46
```

4) Copy one array to another using pointers.(B)

Program :-

```
#include <stdio.h>
void main(){
    int size = 0,i=0;
    printf("Enter the size Of Array : ");
    scanf("%d",&size);
    int src[size];
    for(i=0;i<size;i++){
        printf("Enter the Number at idx %d :",i);
        scanf("%d",&src[i]);
    }
    int copy[size];
    int *srcPtr = src;
    int *copyPtr = copy;
    /*(arr+1) = arr[1]
    for(i=0;i<size;i++){
        *(copyPtr+i) = *(srcPtr+i);
    }
    printf("\nCopy Array : ");
    for(i=0;i<size;i++){
        printf("%d ",copy[i]);
    }
}
```

Output :-

```
Enter the size Of Array : 5
Enter the Number at idx 0 :12
Enter the Number at idx 1 :34
Enter the Number at idx 2 :56
Enter the Number at idx 3 :23
Enter the Number at idx 4 :45

Copy Array : 12  34  56  23  45
```

5) Swap two arrays using pointers.(C)

Program :-

```
#include <stdio.h>
void main(){
    int size = 0,i=0;
    printf("Enter the size Of Array : ");
    scanf("%d",&size);

    //Array1
    int arr1[size];
    printf("\nEnter the value in Array1 : \n");
    for(i=0;i<size;i++){
        printf("Enter the Number at idx %d :",i);
        scanf("%d",&arr1[i]);
    }

    //Array2
    int arr2[size];
    printf("\nEnter the value in Array2 : \n");
    for(i=0;i<size;i++){
        printf("Enter the Number at idx %d :",i);
        scanf("%d",&arr2[i]);
    }

    //swap using pointer
    int *ptr1 = arr1;
    int *ptr2 = arr2;
    int temp = 0;
    for(i=0;i<size;i++){
        temp = *(ptr2+i);
        *(ptr2+i) = *(ptr1+i);
        *(ptr1+i) = temp;
    }

    //Print Array1
    printf("\nArray1 : \n");
    for(i=0;i<size;i++){
        printf("%d ",arr1[i]);
    }

    //Print Array2
    printf("\nArray2 : \n");
    for(i=0;i<size;i++){
        printf("%d ",arr2[i]);
    }
}
```



Output :-

```
Enter the size Of Array : 5

Enter the value in Array1 :
Enter the Number at idx 0 :1
Enter the Number at idx 1 :2
Enter the Number at idx 2 :3
Enter the Number at idx 3 :4
Enter the Number at idx 4 :5

Enter the value in Array2 :
Enter the Number at idx 0 :6
Enter the Number at idx 1 :7
Enter the Number at idx 2 :8
Enter the Number at idx 3 :9
Enter the Number at idx 4 :10

Array1 :
6 7 8 9 10
Array2 :
1 2 3 4 5
```